

An Empirical Study on Tonal Fidelity Preservation for Speech Enhancement Model Selection in Vietnamese

Abstract— Modern speech enhancement has achieved promising results, yet most models are developed and evaluated on English-language corpora, raising concerns about their applicability to tonal languages such as Vietnamese, where F_0 contours carry lexical meaning. This study empirically evaluates ten speech enhancement models to investigate (1) their zero-shot performance on Vietnamese and (2) whether in-domain fine-tuning improves perceptual quality and tonal preservation.

Ten representative architectures spanning sub-band, state-space, attention/conformer, and generative paradigms are evaluated on a controlled Vietnamese corpus of 9,400 utterance pairs using PESQ, STOI, F_0 -RMSE, and Pitch Failure Rate (PFR). Results show architecture-dependent differences in adaptation. Sub-band models exhibit reduced pitch fidelity after fine-tuning, whereas the evaluated attention/conformer models generally improve, suggesting that architectural design may influence tonal preservation. MP-SENet achieves the most balanced performance, reaching a PESQ of 3.611, STOI of 0.971, F_0 -RMSE of 11.27 Hz, and 0% PFR after fine-tuning. Its advantage is maintained across the observed checkpoints, with no clear PESQ plateau within the evaluated training horizon. In contrast, MetricGAN+ demonstrates a mismatch between perceptual optimization and tonal fidelity: despite improved perceptual performance after adaptation, its pitch error increases from 202 Hz to 243 Hz.

Overall, the results show that in-domain fine-tuning alone does not guarantee tonal preservation and that conventional perceptual metrics may overlook linguistically relevant pitch distortion. These findings motivate the joint use of pitch-aware and perceptual metrics when evaluating speech enhancement systems for Vietnamese and other tonal languages.

Index Terms—Speech Enhancement, Vietnamese Speech, Deep Learning, Speech Processing, PESQ, STOI, F_0 -RMSE

I. INTRODUCTION

IN recent years, speech enhancement (SE) has advanced rapidly through deep learning, with architectures such as SEMamba [1], FullSubNet [2], and MetricGAN+ [3] achieving substantial improvements in noise suppression and speech quality. However, their cross-lingual generalization remains insufficiently studied, as most models are developed and evaluated on English-centric corpora.

Although the source-filter model [4] suggests that speech enhancement is largely independent of linguistic content, data-driven models may encode language-specific acoustic statistics. This poses a particular challenge for Vietnamese, where fundamental-frequency (F_0) contours carry lexical information [5]. Consequently, models optimized for non-tonal speech may suppress or distort pitch while still achieving high conventional perceptual scores. This motivates evaluating Vietnamese speech enhancement jointly in terms of perceptual quality, intelligibility, and tonal preservation.

The main contributions of this work are:

- We systematically evaluate ten speech enhancement architectures on Vietnamese speech using a joint PESQ, STOI, F_0 -RMSE, and Pitch Failure Rate (PFR) protocol. The dataset and code are publicly available at <https://github.com/hkha0801-sketch/PESEM-VS.git> [6].
- We incorporate F_0 -RMSE and PFR as complementary pitch-preservation metrics, revealing tonal degradation that may be overlooked by conventional PESQ/STOI evaluation.
- Through multi-criteria evaluation and in-domain fine-tuning, we identify a balanced architecture for Vietnamese speech enhancement and assess how adaptation affects its perceptual and tonal performance.

The remainder of this paper reviews related architectures in Section II, describes the methodology in Section III, presents the results in Section IV, and discusses the findings and conclusions in Section V.

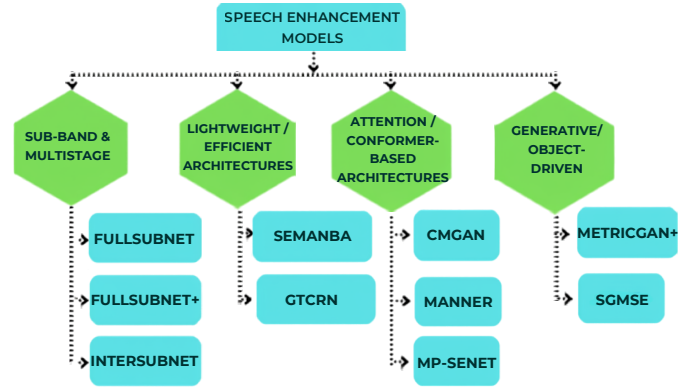


Fig. 1. Taxonomy of speech enhancement paradigms.

II. RELATED WORK

This section provides an overview of relevant studies.

A. Lightweight and Resource-Efficient Architectures

Lightweight architectures prioritize real-time enhancement with low computational cost. SEMamba [1] uses Selective State Space Models for efficient long-range temporal modeling, while GTCRN [7] combines grouped convolutions, recurrent modeling, and sub-band features with a small parameter footprint. Their compact representations motivate evaluating whether computational efficiency compromises the preservation of fine-grained F_0 information in Vietnamese speech.

B. Sub-band and Multi-stage Processing

Sub-band architectures model local frequency regions while retaining global spectral context. FullSubNet [2], FullSubNet+ [8], and InterSubNet [9] progressively extend this principle through full/sub-band fusion, channel attention, and inter-subband interaction, respectively. Such frequency-local modeling is particularly relevant to Vietnamese, where lexical tones depend on fine-grained F_0 trajectories, motivating evaluation of their pitch-preservation capability.

C. Attention and Conformer-based Architectures

Attention-based architectures capture global time-frequency dependencies without fixed sub-band decomposition. MAN-NER [10] employs multi-view attention, while CMGAN [11] uses conformer-based temporal-spectral modeling. MP-SENet [12] further estimates magnitude and phase separately using a conformer backbone. Their global modeling capacity makes them important candidates for assessing whether long-range spectral representations better preserve Vietnamese tonal contours.

D. Generative and Metric-Driven Speech Enhancement

Metric-driven and generative approaches optimize enhancement beyond conventional spectral regression. MetricGAN+ [3] uses adversarial learning to directly target perceptual metrics such as PESQ, whereas SGMSE [13] reconstructs clean speech through score-based generative denoising. Since these objectives are not explicitly designed for tonal preservation, their inclusion tests whether perceptual or generative optimization also preserves Vietnamese F_0 information.

E. Summary and Research Gap

Across these directions — lightweight and resource-efficient design, sub-band decomposition, attention/conformer-based global modeling, and generative or metric-driven optimization — speech enhancement research has diversified substantially in signal representation, architecture design, and training objectives. However, many of these architectures have been developed and evaluated primarily on English-centric corpora. Because modern data-driven models implicitly learn language-specific statistical representations, applying them across languages creates an out-of-distribution condition in which pitch dynamics may be misinterpreted as noise. Consequently, standard noise-suppression evaluations may not fully capture the preservation of language-specific acoustic structures, particularly the F_0 contours essential to Vietnamese lexical tones. This work addresses this gap through a systematic empirical study, evaluating ten representative architectures across these four categories under a unified protocol that combines standard quality metrics with pitch-preservation measures (F_0 -RMSE and PFR) to compare how the evaluated architectures balance perceptual quality, intelligibility, and tonal fidelity for Vietnamese speech.

III. METHODOLOGY

This section describes the proposed methodology, including model selection, dataset construction, evaluation metrics, and experimental design. The overall pipeline is depicted in Fig. 2.

A. Model Selection Criteria

TABLE I
CANDIDATE ARCHITECTURES CONSIDERED AND SELECTION RATIONALE

Model	Status	Category / Rationale
FullSubNet[2]	Included	Sub-band architecture
FullSubNet+[8]	Included	Sub-band architecture
InterSubNet[9]	Included	Sub-band architecture
SEMamba[1]	Included	Lightweight/resource-efficient architecture
GTCRN[7]	Included	Lightweight/resource-efficient architecture
MANNER[10]	Included	Attention/conformer-based architecture
CMGAN[11]	Included	Attention/conformer-based architecture
MP-SENet[12]	Included	Attention/conformer-based architecture
MetricGAN+[3]	Included	Generative/objective-driven architecture
SGMSE[13]	Included	Generative/objective-driven architecture
Conv-TasNet[14]	Excluded	Criterion (i): designed for multi-speaker separation, not single-source denoising
DeepFilterNet[15]	Excluded	Criterion (ii): trained/optimized for 48 kHz full-band input vs. the 16 kHz pipeline used here
Spiking-FullSubNet[16]	Excluded	Criterion (i): SNN/surrogate-gradient training paradigm incompatible with the unified backpropagation protocol
LiSenNet[17]	Excluded	Criterion (iii): offers no additional insight into F_0 -contour behavior beyond GTCRN's lightweight sub-band/dual-path design
FSPEN[18]	Excluded	Criterion (iii): offers no additional insight into F_0 -contour behavior beyond GTCRN's lightweight sub-band/dual-path design

The ten architectures evaluated in this study were selected from a broader pool of fifteen candidate models under three criteria: (i) task compatibility with single-source, single-channel denoising; (ii) compatibility with the 16 kHz sampling-rate protocol adopted throughout the dataset pipeline; and (iii) relevance to tonal-language pitch preservation analysis, ensuring the four representative categories introduced in Section II are each populated by architecturally distinct models whose design choices are informative for diagnosing F_0 - contour behavior under Vietnamese tonal speech — rather than near-duplicate variants that would offer no additional insight into this study's central question. Table I lists all fifteen candidates together with their inclusion status and the corresponding rationale.

B. Dataset Construction

The evaluation dataset comprises a total of 9,400 noisy-clean paired speech samples, created by mixing clean Vietnamese speech with background noise. Specifically, the core

deficits observed in Experiment 1. Results are presented in Section IV-B.

- **Experiment 3 — Multi-Criteria Model Selection:** PESQ, STOI, F_0 -RMSE, and PFR are jointly considered to identify architectures that balance perceptual quality, intelligibility, and tonal preservation. Results are presented in Section IV-C.
- **Experiment 4 — Checkpoint Trajectory Analysis:** The leading candidates from Experiment 3 are analyzed across fine-tuning checkpoints to examine convergence and the evolution of perceptual and pitch-preservation performance. Results are presented in Section IV-D.

IV. RESULTS

This section reports the outcomes of the four experiments defined in Section III-E, progressing from zero-shot evaluation and in-domain adaptation to multi-criteria model selection and checkpoint-level analysis.

A. Zero-Shot Cross-Lingual Evaluation

Following Experiment 1 in Section III-E, we first evaluate the cross-lingual generalizability of English-pretrained models on the Vietnamese test set without adaptation. As reported in Table III, the baseline results reveal a pronounced performance disparity across architectural classes. While standard non-intrusive metrics indicate viable acoustic noise suppression for several models, incorporating tonal fidelity metrics immediately exposes severe cross-lingual mismatch.

TABLE III
VIETNAMESE SPEECH ENHANCEMENT RESULTS.

Model	PESQ	STOI	F_0 -RMSE (mean, Hz)	PFR
FullSubNet	2.643	0.938	30.87	2.66%
FullSubNet+	2.749	0.947	31.92	0.64%
InterSubNet	2.863	0.949	28.09	0.74%
SEMamba	1.921	0.902	67.22	1.70%
GTCRN	2.133	0.921	41.16	1.81%
MANNER	1.203	0.663	128.46	5.74%
CMGAN	2.011	0.916	55.23	1.60%
MP-SENet	3.283	0.960	20.89	1.38%
MetricGAN+	1.715	0.801	202.08	5.74%
SGMSE	1.644	0.891	53.84	3.62%

MP-SENet achieves the strongest zero-shot performance among the evaluated models, attaining a PESQ of 3.283, a STOI of 0.960, and an F_0 -RMSE (Eq. (2)) of 20.89 Hz. Sub-band models achieve PESQ scores of 2.643–2.863 with F_0 -RMSE values of 28.09–31.92 Hz. In contrast, MetricGAN+ and MANNER show substantially larger pitch errors. MetricGAN+ reaches an F_0 -RMSE of 202.08 Hz and a PFR of 5.74%, while MANNER obtains a PESQ of 1.203 and an F_0 -RMSE of 128.46 Hz.

These zero-shot findings indicate that satisfactory acoustic enhancement performance does not necessarily imply reliable cross-lingual pitch preservation. This motivates examining

whether domain-specific fine-tuning improves pitch preservation under the evaluated conditions.

B. In-Domain Fine-Tuning Performance

Following Experiment 2 in Section III-E, all ten models are fine-tuned on the Vietnamese corpus to determine whether in-domain adaptation mitigates the cross-lingual deficits observed in Experiment 1. As shown in Table IV, the effects vary substantially across architectural classes, indicating that adaptation benefits are not uniform across model designs.

Several attention/conformer and state-space models show notable improvements after adaptation. Among the evaluated models, MP-SENet achieves the strongest overall metric combination (PESQ 3.611, STOI 0.971, F_0 -RMSE 11.27 Hz, PFR 0.00%), while CMGAN and SEMamba also show substantial improvements in both perceptual quality and pitch preservation. Notably, CMGAN reduces its F_0 -RMSE from 55.23 Hz to 15.27 Hz, whereas SEMamba reduces it from 67.22 Hz to 16.90 Hz, indicating improved pitch preservation after Vietnamese fine-tuning. SGMSE similarly improves to 16.82 Hz F_0 -RMSE with 0.00% PFR. In contrast, MetricGAN+ remains severely pitch-distorted (243.44 Hz) despite a modest PESQ improvement, highlighting a mismatch between perceptual optimization and tonal fidelity.

Sub-band models show limited or negative transfer. FullSubNet and InterSubNet degrade in both PESQ and F_0 -RMSE, while FullSubNet+ remains perceptually stable but exhibits increased pitch error. This contrast suggests an association between broader temporal-spectral modeling and more favorable adaptation behavior in the evaluated models. Overall, these results indicate that in-domain data alone does not guarantee tonal preservation; adaptation outcomes vary substantially across the evaluated architectures.

C. Multi-Criteria Model Selection

TABLE V
MULTI-CRITERIA ARCHITECTURE TIERING

Tier	Model	PESQ	STOI	F_0 -RMSE	PFR	Observed Characteristics
1	MP-SENet	High	High	Superior	Low	Strong across all dimensions
	CMGAN	High	High	High	Superior	Strong fidelity, low failure rate
2	SEMamba	High	High	Moderate	Low	Competitive overall performance
	SGMSE	Moderate	High	Moderate	Superior	Zero PFR, moderate PESQ
3	GTCRN	Moderate	Moderate	Moderate	Low	Moderate overall performance
	FullSubNet+	Moderate	High	Poor	Low	Higher pitch error
	FullSubNet	Moderate	Moderate	Poor	Low	Degradation after fine-tuning
	MetricGAN+	Poor	Poor	Degraded	Poor	Severe pitch degradation

Following Experiment 3 in Section III-E, we jointly consider PESQ, STOI, F_0 -RMSE, and PFR to compare architectures in terms of perceptual quality, intelligibility, and tonal preservation. Rather than using a weighted aggregate score,

TABLE IV
SPEECH ENHANCEMENT RESULTS FOR VIETNAMESE SPEECH AFTER IN-DOMAIN FINE-TUNING.

Method	Quality & Intelligibility		Pitch Preservation		
	PESQ	STOI	F0-RMSE (Median)	F0-RMSE (Mean)	PFR
FullSubNet[2]	2.241	0.926	4.69	42.80	0.64%
FullSubNet+[8]	2.726	0.946	3.43	37.14	0.43%
InterSubNet[9]	1.805	0.906	6.91	70.38	0.74%
SEMamba[1]	3.354	0.964	2.99	16.90	0.32%
GTCRN[7]	2.548	0.939	3.71	30.49	0.21%
MANNER[10]	2.212	0.921	3.91	28.75	0.11%
CMGAN[11]	3.393	0.965	2.40	15.27	0.11%
MP-SENet[12]	3.611	0.971	1.97	11.27	0.00%
MetricGAN+[3]	2.050	0.851	248.95	243.44	1.81%
SGMSE[13]	2.943	0.955	2.45	16.82	0.00%

models are qualitatively grouped according to their relative performance across the four metrics. Table V summarizes this multi-criteria tiering.

Tier 1 includes models that remain consistently strong across both perceptual and pitch-related metrics. Tier 2 contains models that are competitive overall but show a noticeable limitation in at least one dimension. Tier 3 includes models with larger trade-offs or degradation in perceptual quality, intelligibility, or pitch preservation. The tiers therefore represent relative multi-metric performance rather than fixed numerical thresholds, preventing a strong result on one metric from compensating for weak tonal preservation.

Under this criterion, MP-SENet and CMGAN provide the most consistent balance across the evaluated metrics and are selected for the checkpoint-level analysis in Experiment 4.

D. Checkpoint Trajectory Analysis

Following Experiment 4 in Section III-E, we analyze the checkpoint trajectories of the two leading candidates identified in Experiment 3, MP-SENet and CMGAN. Fig. 3 compares their PESQ and F0-RMSE across all 20 fine-tuning checkpoints. **CMGAN’s PESQ rises steeply through epoch ~14 then flattens**, oscillating within a narrow 3.39–3.42 band for the remaining checkpoints, while its F0-RMSE continues to fall in the second half of training (55.23 Hz zero-shot → 15.27 Hz at checkpoint 20, a −72.4% reduction) — indicating limited additional PESQ improvement within the observed training horizon while pitch error continues to decrease. MP-SENet’s PESQ increases through checkpoint 19 (3.283 zero-shot → 3.611 at checkpoint 19, +10.0%), with no comparable plateau, and its F0-RMSE remains 4–9 Hz below CMGAN’s across all evaluated checkpoints (20.89 Hz zero-shot → 11.27 Hz, −46.1%).

Although CMGAN exhibits larger relative improvements during fine-tuning, this should be interpreted in light of its weaker zero-shot baseline (PESQ 2.011, F0-RMSE 55.23 Hz), which provides greater room for recovery. MP-SENet, in contrast, starts from a stronger baseline. Across the observed

checkpoints, MP-SENet maintains better absolute PESQ and F_0 -RMSE than CMGAN, while showing no clear PESQ plateau within the evaluated training horizon. These trajectories further support MP-SENet as the most balanced architecture among the models evaluated in this study, reinforcing its selection in Section V.

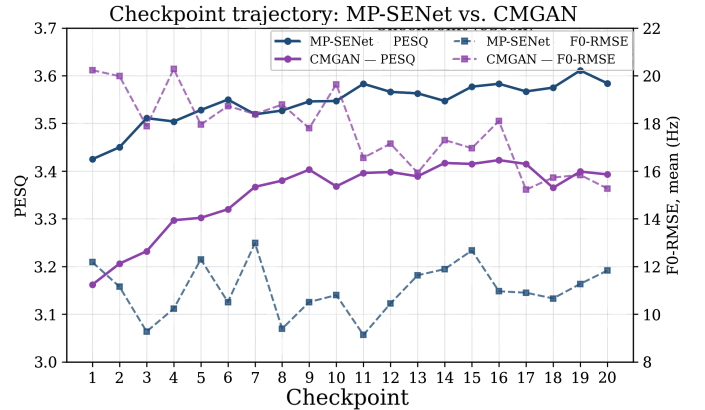


Fig. 3. Checkpoint trajectories of PESQ (solid) and F0-RMSE (dashed) for the two Tier-1 architectures.

V. DISCUSSION & CONCLUSION

A. Summary of Findings and Implications

This study addressed a key limitation in current speech enhancement practice: models developed primarily on English, non-tonal speech are commonly evaluated using PESQ and STOI, which do not explicitly measure preservation of pitch information. For Vietnamese, where F0 contours carry lexical meaning, perceptually acceptable enhancement may therefore still introduce linguistically relevant tonal distortion.

To address this issue, we evaluated ten speech enhancement architectures on Vietnamese speech using a joint PESQ, STOI, F0-RMSE, and PFR protocol. The experiments revealed three main findings. First, English-pretrained models retain useful zero-shot denoising capability, but favorable PESQ/STOI

scores may coexist with substantial pitch degradation. Second, fine-tuning outcomes vary across the evaluated architectures: MP-SENet, CMGAN, and SEMamba improve across perceptual and pitch-related metrics, whereas sub-band models show limited or negative transfer. Thus, in-domain adaptation alone does not guarantee tonal preservation. Third, multi-criteria evaluation identifies MP-SENet and CMGAN as the strongest balanced candidates, with checkpoint analysis further supporting MP-SENet through consistently lower F0-RMSE and continued PESQ improvement.

These findings have two practical implications. First, speech enhancement systems for Vietnamese and other tonal languages should be evaluated with pitch-aware measures such as F0-RMSE and PFR alongside conventional PESQ/STOI metrics. Second, model selection should account for an architecture's ability to preserve tonal information before domain adaptation, since fine-tuning does not consistently improve tonal preservation across the evaluated architectures.

B. Limitations

Despite establishing a controlled multi-accent benchmark with 9,400 samples, several limitations remain. First, evaluation is restricted to a fixed 10 dB SNR, leaving more severe and non-stationary noise conditions unexplored. Second, the corpus contains only 20 core sentences, limiting broader linguistic generalization. Finally, pYIN-based evaluation considers detected voiced segments only, so frame-level F_0 -RMSE may not fully capture distortions near voicing boundaries.

C. Future Work

Future work will explore tonal-aware speech enhancement architectures that combine the phase-aware modeling of MP-SENet [12] with efficient state-space backbones such as SE-Mamba [1]. We will also evaluate tonal preservation under standardized benchmarks, unseen noise, and lower-SNR conditions. Finally, enhanced speech will be integrated with Vietnamese ASR systems such as PhoWhisper [24] to assess whether improved F0 preservation reduces lexical confusion and Word Error Rate (WER).

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